

Table X.A: Conceptual Framework - Expected Effects of the Algorithm on Audit process and Audit Outcomes

Determinants of audit outcome / success/ yield				
Steps in the audit process	Riskiness of case (true evasion)	Inspector effort	Corruption	Inspector bias
Selection into annual program	A increases or lowers average riskiness of cases depending on i) quality of data, ii) value of inspectors' private information, iii) quality of inspectors' mental processing model	A reduces time spent on selection (hence more time available for implementation)	A eliminates corruption	A replaces human bias with bias due to data limitations, ex-ante unclear which drives bigger wedge between optimal and actual selection
Audit start	A reduces likelihood of audit start if inspectors are more likely to start cases they are familiar with	A might reduce likelihood of starting case if inspectors lack information on risks	A can increase or decrease corruption	No change in bias (only direct change due to change in pool of cases)
Audit implementation	A lowers implementation effectiveness if inspectors' knowledge of case is relevant	A reduces inspector autonomy, and could hence reduce motivation and effort during implementation	A can increase or decrease corruption	No change in bias (only direct change due to change in pool of cases)

Table X.B: Realized Effects of the Algorithm on Audit process and Audit Outcomes

Determinants of audit outcome / success/ yield				
Steps in the audit process	Riskiness of case (true evasion)	Inspector effort	Corruption	Inspector bias
Selection into annual program	A cases are less risky than others	Inspectors have more time for implementation, but likely invest this time in inspector-selected cases	No evidence for change in corruption (combining corruption at different stages in audit process)	Data-based biases (errors/noise) are more important than human biases in driving a wedge between optimal and actual selection and audit implementation
Audit start	A cases less likely to be implemented, due to lower riskiness; no evidence that lack of familiarity with a case affects audit start	No evidence that lack of information about A cases or A case risks explains lower implementation		
Audit implementation	A cases have lower yield conditional on implementation, given lower riskiness	No evidence for reduction in effort conditional on audit risk		